# Capacity Plan

**Ref:** 5.7
**Product:** iConnect Plus
**Organisation:** Cache Digitech
**Classification:** Internal — Confidential
**Version:** 2.0
**Date:** 2026-11-03
**Deployment model:** On-premises (Cache Digitech premises) · VPN for remote access
**Audience:** IT infrastructure, platform engineering, management

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## 1. Purpose

Plan **compute, storage, and network capacity** for iConnect Plus on the **single-host (or primary + DR) Docker Compose** deployment at Cache Digitech premises.

## 2. Current baseline (template)

| Resource | Documented value | Source |
|---|---|---|
| Host | SRV-ERP-01 | 1.5 |
| Example IP | `172.16.200.30` | 1.5 |
| vCPU | **Template / Pending sign-off** | IT asset record |
| RAM | **Template / Pending sign-off** | IT asset record |
| Disk (app + data) | **Template / Pending sign-off** | IT |
| Concurrent users (peak) | **Template / Pending sign-off** | Business |
| Tenants | **Template / Pending sign-off** | |

## 3. Growth assumptions (12 months)

| Dimension | Assumption | Trigger to revisit |
|---|---|---|
| Users | **Template / Pending sign-off** % growth | >70% sustained CPU |
| Storage (PostgreSQL) | **Template / Pending sign-off** GB/year | Disk >75% |
| Storage (MinIO) | Attachment growth per module | Disk >75% |
| Celery throughput | Outbox + notification volume | Queue depth sustained |

## 4. Component sizing guidelines

| Component | Sizing note |
|---|---|
| PostgreSQL | Largest consumer — SSD required; monitor `erp_pgdata` |
| MinIO | Plan 2–3× current object size for headroom |
| Redis | Memory ∝ cache keys; session load moderate |
| RabbitMQ | Disk for persistent queues if used |
| API | Scale vertically first (more CPU/RAM to host) |

| | |
|---|---|
| Celery workers | Add worker container or `--concurrency` before second host |
| nginx | Low footprint |
| ClamAV | RAM for defs; CPU on scan spikes |

**Horizontal second app host:** Evolution-gated — not default monolith split.

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## 5. Monitoring thresholds

| Metric | Warning | Critical | Action |
|---|---|---|---|
| CPU sustained | >70% 15 min | >90% 5 min | Investigate queries / workers |
| RAM | >80% | >95% | Restart leak suspect; add RAM |
| Disk `/var/lib/docker` | >75% | >90% | 5.6 disk SOP |
| PostgreSQL connections | **Template / Pending sign-off** | **Template / Pending sign-off** | Pool tuning |
| RabbitMQ queue depth | **Template / Pending sign-off** | **Template / Pending sign-off** | Scale workers |

**Baseline metrics capture: Template / Pending sign-off**

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## 6. Network capacity

| Path | Consideration |
|---|---|
| LAN users | Switch bandwidth to host |
| VPN users | VPN concentrator throughput |
| Backup to NAS | Off-peak scheduling (4.4) |

No public CDN — static assets served from `erp-web` on LAN/VPN.

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## 7. Capacity test plan

| Test | Method | Last run |
|---|---|---|
| Load test API | **Template / Pending sign-off** tool | **Pending** |
| Upload burst | Large file to MinIO + ClamAV | **Pending** |
| DB growth projection | Table size query monthly | **Pending** |

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## 8. Upgrade path

| Stage | Action |
|---|---|
| 1 | Vertical — RAM/CPU/disk expand on SRV-ERP-01 |
| 2 | Split infra — DB on dedicated host (same premises) |
| 3 | DR host active-passive (5.5) |
| 4 | Optional OpenSearch cluster — gated |

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## 9. Related documents

| Ref | Document |
|---|---|

| | |
|---|---|
| 1.5 | Asset & Component Inventory |
| 5.1 | Logging & Monitoring |
| 5.6 | Operations Runbook |
| 1.1 | Solution Architecture |

## 10. Change log

| Date | Change | By |
|---|---|---|
| 2026-11-03 | v2.0 capacity template | Platform |

**Review cycle:** Annual — **next review Template / Pending sign-off**